

BIM Expression Derivation

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1 Deduction of Relevance Formula for BIM

$$\begin{aligned} P(R = 1|x, q) &= \frac{P(R = 1 \cap x \cap q)}{P(x \cap q)} = \frac{P(R = 1 \cap x \cap q)}{P(R = 1 \cap q)} * \frac{P(R = 1 \cap q)}{P(x \cap q)} \\ &= P(x|R = 1, q) * \frac{\frac{P(R=1 \cap q)}{P(q)}}{\frac{P(x \cap q)}{P(q)}} = \frac{P(x|R = 1, q) x P(R = 1|q)}{P(x|q)} \end{aligned}$$

2 Deduction of Non-Relevance Formula for BIM

$$\begin{aligned} P(R = 0|x, q) &= \frac{P(R = 0 \cap x \cap q)}{P(x \cap q)} = \frac{P(R = 0 \cap x \cap q)}{P(R = 0 \cap q)} * \frac{P(R = 0 \cap q)}{P(x \cap q)} \\ &= P(x|R = 0, q) * \frac{\frac{P(R=0 \cap q)}{P(q)}}{\frac{P(x \cap q)}{P(q)}} = \frac{P(x|R = 0, q) x P(R = 0|q)}{P(x|q)} \end{aligned}$$

3 Deduction of Odds of Relevance

$$O(R|x, q) = \frac{P(R = 1|x, q)}{P(R = 0|x, q)} = \frac{\frac{P(x|R=1, q) P(R=1|q)}{P(x|q)}}{\frac{P(x|R, q) P(R=0|q)}{P(x|q)}} = \frac{P(R = 1|q)}{P(R = 0|q)} \frac{P(x|R = 1, q)}{P(x|R = 0, q)}$$